DL26, Assignment 1

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Task 2a)

For the given FNN with m sigmoid neurons at the hidden layer:

$$z_j = \sigma(w_{1j}x + b_{1j}), \qquad j = 1, \dots, m,$$

and

$$\hat{y}(x) = \sum_{j=1}^{m} w_{2j} z_j + b_2.$$

Substituting the hidden activations into the output gives

$$\hat{y}(x) = \Big(\sum_{j=1}^{m} w_{2j} \cdot \sigma(w_{1j}x + b_{1j}) \Big) + b_2.$$

Thus, the network output is a weighted sum of sigmoid functions. Positive weights contribute upward slopes, negative weights downward slopes. Hence, one positive/negative pair can form one peak. The given dataset is a sine wave over two periods. The fit for a given m may look like:

(i) For $m = 0$, the model reduces to linear regression, i.e. a straight line and a poor fit.

(ii) For $m = 1$, the graph is one sigmoid-shaped curve which cannot capture the oscillation of a sine wave.

(iii) For $m = 2$, one peak or a trough can be represented but still not the full oscillation.

(iv) For $m = 3$, one peak plus one additional upward or downward slope can be represented which is maybe 1.5 times a period of the sine wave.

Task 2b)

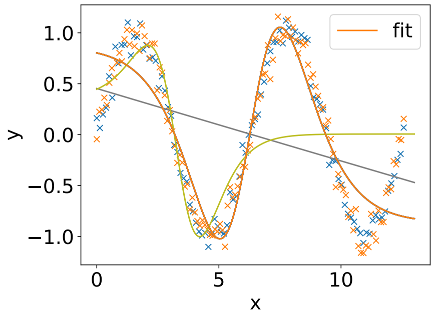

Figure 1: Repeated training runs for MLP function with two hidden neurons.

(i) Compared with the conjecture from 2a, the fitted curve for $m = 2$ shows not only a peak, but also a trough, and it becomes approximately horizontal for large positive or negative x. The model is not able to fit our data. The values of MSE obtained for both Training and Test data are around 0.3 which validate our conjecture and the conclusion.

(ii) Given training data $\{(x_n, y_n)\}_{n=1}^{N}$, the objective (cost) function is the MSE

$$J = \frac{1}{N} \sum_{n=1}^{N} \left[y_n - (w_{21}\,\sigma(w_{11}x_n + b_{11}) + w_{22}\,\sigma(w_{12}x_n + b_{12}) + b_2)\right]^2 .$$

Since this objective function is non-convex, different random initializations of the MLP function can lead BFGS to different local optima and thus to different fitted curves. BFGS itself is deterministic once the initialization is fixed.
One of the fits also appears almost linear, because the sigmoid is approximately linear near 0. If the pre-activations stay close to 0, the weighted sum of the two sigmoid units can behave almost like a straight line.

Task 2c)

Table 1: Training and test MSE for different hidden-layer widths.

Width m	Training MSE	Test MSE
1	0.3729	0.3743
2	0.0796	0.0867
3	**0.0073**	**0.0103**
10	0.0053	2.4624
50	0.0024	1.7884
100	0.0014	6.7874

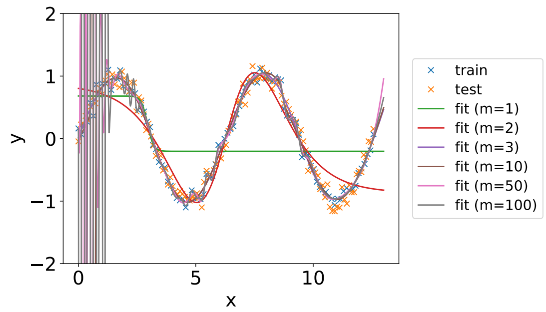

Figure 2: Fits of the MLP for different hidden-layer widths.

(i) As the value of m increases, i.e. for $m = 10, 50$, and 100, the training error decreases further, but the test error increases strongly, which indicates overfitting. The irregular behavior near the left boundary can be explained by the high flexibility of the model or steep sigmoid transitions caused by large learned weights.
The best hidden-layer width is $\mathbf{m} = \mathbf{3}$, since it gives the lowest test error while still fitting the training data very well.

(ii) The results are consistent with the expectation from 2a that more hidden neurons increase the expressivity of the network. For $m = 1$, the model underfits, while $m = 2$ and especially $m = 3$ captures the sine-shaped structure much better.

Task 2d)

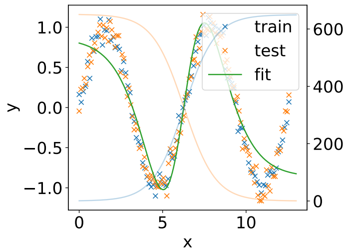

Figure 3: $m = 2$

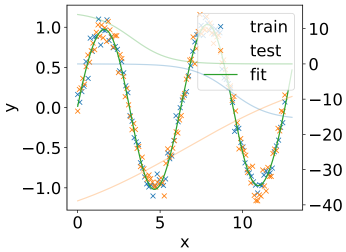

Figure 4: $m = 3$

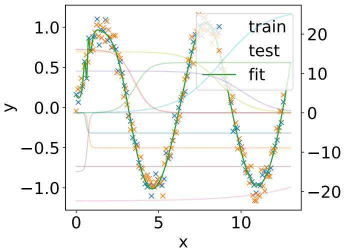

Figure 5: $m = 10$

(i) and (ii) For two hidden neurons, it is convenient to write the scaled components as

$$s_j(x) = w_{2j}\,\sigma(w_{1j}x + b_{1j}) = \frac{w_{2j}}{1 + e^{-(w_{1j}x+b_{1j})}}, \qquad j = 1, 2,$$

so that the final output is

$$\hat{y}(x) = s_1(x) + s_2(x) + b_2.$$

Using this scaled sigmoid representation, we can see how the parameters affect the individual sigmoid functions as components, and their combination as the final output:

- w_{1j}: direction and steepness of component $s_j(x)$; thus how sharply this part changes in the final output
- b_{1j}: horizontal position of component $s_j(x)$; thus where this part appears in the final output
- w_{2j}: sign and vertical scaling of component $s_j(x)$; thus whether it contributes upward or downward, and how strongly, to the final output
- b_2: no effect on the individual components; vertical shift of the final output

As shown in Figure 3, for $m = 2$ the fitted curve is obtained by summing two scaled sigmoid components and the output bias. This representation is still fairly intuitive, since each component can be examined separately and its contribution to the final output can be understood directly. Hence, the flexibility of the FNN arises from combining scaled sigmoids with different positions, slopes, directions, and magnitudes.

(iii) For $m = 3$ and $m = 10$, the individual components are interpreted in the same way, since they are still scaled sigmoids. However, comparing Figures 4 and 5 with Figure 3, the overall representation becomes harder to interpret as more components overlap. In particular, $m = 10$ is much less transparent than $m = 2$.

Task 2e)

Task 3a)

For one scalar input $x \in \mathbb{R}$ and $W_1 \in \mathbb{R}^{1\times 50}$, $b_1 \in \mathbb{R}^{50\times 1}$, $W_2 \in \mathbb{R}^{50\times 1}$, $b_2 \in \mathbb{R}$, the forward pass is

1. $z_1 = W_1^\top x \in \mathbb{R}^{50\times 1}$
2. $z_2 = z_1 + b_1 \in \mathbb{R}^{50\times 1}$
3. $z_3 = \sigma(z_2) \in \mathbb{R}^{50\times 1}$, where $[z_3]_k = \frac{1}{1+e^{-[z_2]_k}}$
4. $z_4 = W_2^\top z_3 \in \mathbb{R}$
5. $\hat{y} = z_4 + b_2 \in \mathbb{R}$
6. $l = L(y, \hat{y}) = (y - \hat{y})^2 \in \mathbb{R}$

Task 3b)

We compute $\delta_v = \frac{\partial l}{\partial v}$ for each node v in reverse topological order, applying the chain rule at each step.

1. Loss node: $\delta_l = \frac{\partial l}{\partial l} = 1$

2. Through $l = (y - \hat{y})^2$**:**

$\delta_{\hat{y}} = \frac{\partial l}{\partial \hat{y}} = 2(\hat{y} - y) \cdot \delta_l = 2(\hat{y} - y) \in \mathbb{R}$

$\delta_y = \frac{\partial l}{\partial y} = 2(y - \hat{y}) \cdot \delta_l = -\delta_{\hat{y}} \in \mathbb{R}$

3. Through $\hat{y} = z_4 + b_2$ **:**

$\delta_{z_4} = \delta_{\hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_4} = \delta_{\hat{y}} \cdot 1 = \delta_{\hat{y}} \in \mathbb{R}$

$\delta_{b_2} = \delta_{\hat{y}} \cdot \frac{\partial \hat{y}}{\partial b_2} = \delta_{\hat{y}} \cdot 1 = \delta_{\hat{y}} \in \mathbb{R}$

4. Through $z_4 = W_2^\top z_3$ **:**

$\delta_{z_3} = W_2 \, \delta_{z_4} \in \mathbb{R}^{50\times1}$

$\delta_{W_2} = z_3 \, \delta_{z_4}^\top \in \mathbb{R}^{50\times1}$

5. Through $z_3 = \sigma(z_2)$ **:**

$\delta_{z_2} = \delta_{z_3} \odot \sigma'(z_2) = \delta_{z_3} \odot z_3 \odot (1 - z_3) \in \mathbb{R}^{50\times1}$

since $\sigma'(a) = \sigma(a)(1 - \sigma(a))$ and $z_3 = \sigma(z_2)$.

6. Through $z_2 = z_1 + b_1$ **:**

$\delta_{z_1} = \delta_{z_2} \cdot 1 = \delta_{z_2} \in \mathbb{R}^{50\times1}$

$\delta_{b_1} = \delta_{z_2} \cdot 1 = \delta_{z_2} \in \mathbb{R}^{50\times1}$

7. Through $z_1 = W_1^\top x$ **:**

$\delta_{W_1} = x \, \delta_{z_1}^\top \in \mathbb{R}^{1\times50}$

$\delta_x = W_1 \, \delta_{z_1} \in \mathbb{R}$

Ehrenwörtliche Erklärung

Ich versichere, dass ich die beiliegende Bachelor-, Master-, Seminar-, oder Projektarbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Quellen und in der untenstehenden Tabelle angegebenen Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen. Ich bin mir bewusst, dass eine falsche Erklärung rechtliche Folgen haben wird.

Declaration of Used AI Tools

Task	Solved with AI?	Tool	Purpose	Useful?
1b	Yes	ChatGPT	Generate draft solution	yes
1c	Only Guidance	ChatGPT	Understanding nn.Module and nn.Parameter	Yes
1d				
2a				
2b	Minimal	ChatGPT	Graph Change and wordings improvement	Yes
2c	Minimal	ChatGPT	Latex codes	Yes
2d	Minimal	ChatGPT	Latex codes and wordings improvement	Yes
3a	Yes	ChatGPT	Format latex	yes
3b	Yes	ChatGPT	Format latex, tensor dimension debugging assistance	yes

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